

FracDimPy: A Comprehensive Python Package for Fractal Dimension Calculation and Multifractal Analysis

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Summary

FracDimPy is a comprehensive Python package for fractal dimension calculation and multifractal analysis. Fractal geometry provides powerful tools for characterizing irregular, self-similar structures that appear across scientific disciplines, from geological formations and material surfaces to biological systems and financial markets (Lopes & Betrouni, 2009; Mandelbrot, 1982). FracDimPy offers a unified toolkit that integrates eight monofractal analysis methods, comprehensive multifractal analysis, and built-in fractal generation tools. The package is designed for both research and educational use, featuring an intuitive API, extensive documentation, and rich visualization capabilities built on NumPy, SciPy, Matplotlib, and pandas (Harris et al., 2020; Hunter, 2007; McKinney, 2010; Virtanen et al., 2020).

The monofractal methods include box-counting with configurable boundary handling and partitioning strategies (Foroutan-pour et al., 1999; Liebovitch & Toth, 1989), Hurst exponent via R/S analysis (Hurst, 1951), detrended fluctuation analysis (DFA) (Peng et al., 1994), correlation dimension via the Grassberger-Procaccia algorithm (Grassberger & Procaccia, 1983), information dimension, structural function, variogram, and sandbox methods. The multifractal module provides multifractal DFA (MF-DFA) (Kantelhardt et al., 2002), multifractal spectrum computation for 1D curves and 2D images, and custom epsilon sequences for flexible scaling range optimization. The generator module produces deterministic fractals (Sierpinski triangle, Koch curve, Cantor set, Menger sponge) and stochastic fractals (fractional Brownian motion, Weierstrass-Mandelbrot functions) with known theoretical dimensions, enabling algorithm validation and educational demonstrations.

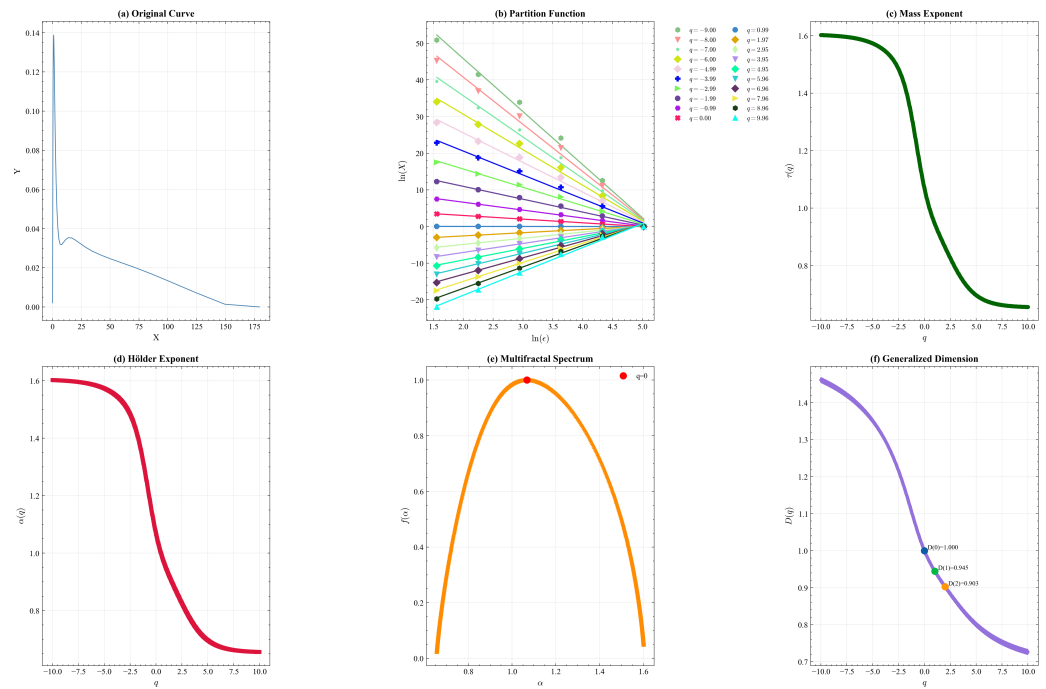


Figure 1: Multifractal analysis of a test signal using FracDimPy: (a) original curve, (b) partition function, (c) mass exponent, (d) Holder exponent, (e) multifractal spectrum, and (f) generalized dimension.

28 Statement of Need

29 Fractal and multifractal analysis has become increasingly important across scientific research,
30 yet practitioners face significant challenges in accessing appropriate computational tools.
31 Existing software is often fragmented across multiple packages with limited methodological
32 coverage, or locked behind commercial licenses with steep learning curves (Lopes & Betrouni,
33 2009).

34 While tools like Nolds (Scholz, 2019), Fathon (Russo & S. Carlo, 2019), and FracLab (INRIA,
35 2020) provide excellent implementations of specific methods, FracDimPy offers a uniquely
36 comprehensive suite that integrates monofractal and multifractal analysis within a single,
37 well-documented package. This unified approach eliminates the need for researchers to learn
38 multiple software environments and ensures consistent data handling across different analysis
39 methods.

40 All implementations have been validated against canonical fractals with known
41 theoretical dimensions—Sierpinski triangle ($D = \log 3 / \log 2 \approx 1.585$), Koch curve
42 ($D = \log 4 / \log 3 \approx 1.262$), and Cantor set ($D = \log 2 / \log 3 \approx 0.631$)—with relative errors
43 consistently below 2%. The package has been applied in geoscience (NMR pore-structure
44 analysis, terrain characterization), materials science (surface roughness quantification), and
45 petroleum engineering (reservoir heterogeneity assessment).

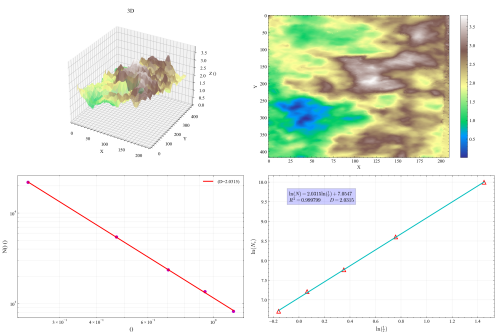


Figure 2: Box-counting analysis of a fractal surface ($D = 2.0315$, $R^2 = 0.9999$), demonstrating the package's capability for 2D surface data.

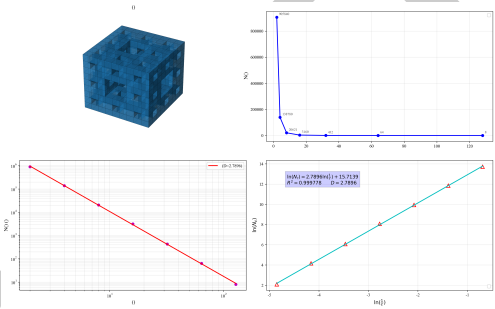


Figure 3: Box-counting analysis of porous media: the log-log plots confirm power-law scaling with high precision.

46 FracDimPy integrates seamlessly with the Python scientific ecosystem, enabling researchers to
47 incorporate fractal analysis into existing NumPy/SciPy/Matplotlib workflows without additional
48 dependencies beyond standard scientific Python packages. The GPL-3.0 license ensures broad
49 accessibility for both academic and commercial applications.

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